MIL-M-0038510/1D(USAF)
1 December 1975
USED IN LIEU OF
MIL-M-38510/1B
23 December 1971
SUPERSEDING
MIL-M-0038510/1C(USAF)
15 October 1973

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, NAND GATES, MONOLITHIC SILICON

This limited coordination military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/1B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, positive NAND logic gating microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/001	02	S	T	D
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Single, 8-input positive NAND gate- -	5430
02	Dual, 4-input positive NAND gate- - -	5420
03	Triple, 3-input positive NAND gate- -	5410
04	Quadruple, 2-input positive NAND gate-	5400
05	Hex, 1-input inverter gate- - - - - -	5404
06	Triple, 3-input positive NAND gate (open collector output)- - - - - - -	5412
07	Quadruple, 2-input positive NAND gate (open collector output)- - - - - - -	5401

1.2.2 Device class. The device class shall be Class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range- - - - - - - - - - - - -	-0.5 Vdc to 7.0 Vdc
Input voltage range - - - - - - - - - - - - -	0 to 5.5 Vdc
Storage temperature range - - - - - - - - - -	-65°C to 150°C
Lead temperature (soldering, 10 seconds)- - -	300°C

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1.4 Recommended operating conditions.

Ambient operating temperature range - - - - - -	-55°C to 125°C
Supply voltage- - - - - - - - - - - - - - - - -	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage- - - - - - - -	2.0 Vdc
Maximum low level input voltage - - - - - - - -	0.8 Vdc

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight, and physical dimensions. The design, construction, weight, and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C, and 125°C. For all tests, the devices shall meet the limits specified in table I.

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TABLE I. Electrical characteristics tests (device type 01).

Test No. 1/ 2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2	-	-	V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V	-	0.8	-	V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.8 V, I_{OH} = -400 µA	2.4	-	±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.8 V, I_{OL} = 16 mA	-	0.4	±.04	V
5*	High level input current (each input)	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.4 V	-	40	±4.0	µA
		I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V	-	100	N/A	µA
6	Low level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IL} = 0.4 V	-0.7	-1.6	-	mA
7	Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IL} = 0 V	-20	-55	-	mA
8	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IH} = 5.5 V	-	6	-	mA

See footnotes at end of this table.

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TABLE I. Electrical characteristics tests (device type 01).-Continued.

Test No. 1/ 2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
9	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IL} = 0 V	-	2	-	mA
10	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_A = 25°C	-	-1.5	-	V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. Electrical characteristics test (device type 02).

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
1*	V_{OL}	2.0 V	16 mA		4.5 V		5.5 V	5.5 V	5.5 V	5.5 V		GND	2.0 V	2.0 V	2.0 V		.4	±.04	V
2*	V_{OL}	5.5 V					2.0 V	2.0 V	2.0 V	2.0 V	16 mA		5.5 V	5.5 V	5.5 V		.4	±.04	
3*	V_{OH}	0.8 V	-.4 mA				5.5 V	5.5 V	5.5 V	5.5 V			5.5 V	5.5 V	5.5 V	2.4		±.12	
4*		5.5 V	-.4 mA				5.5 V						0.8 V	5.5 V	5.5 V				
5*			-.4 mA				5.5 V						5.5 V	0.8 V	5.5 V				
6*			-.4 mA				5.5 V						5.5 V	5.5 V	0.8 V				
7*							0.8 V	5.5 V			-.4 mA		5.5 V	5.5 V	5.5 V				
8*							5.5 V	0.8 V	5.5 V		-.4 mA		5.5 V	5.5 V	5.5 V				
9*							5.5 V	5.5 V	0.8 V	5.5 V	-.4 mA		5.5 V	5.5 V	5.5 V				
10*	V_{OH}	5.5 V			4.5 V		5.5 V	5.5 V	5.5 V	0.8 V	-.4 mA		5.5 V	5.5 V	5.5 V	2.4		±.12	V
11	I_{OS}	GND	GND		5.5 V								GND	GND	GND	-20	-55		mA
12	I_{OS}				5.5 V		GND	GND	GND	GND	GND					-20	-55		mA
13*	I_{IH1}	2.4 V			5.5 V		GND	GND	GND	GND			GND	GND	GND		40	10X or 4 µA	µA
14*	I_{IH1}	GND			5.5 V		GND	GND	GND	GND		GND	2.4 V	GND	GND		40	10X or 4 µA	µA

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 02).--Continued

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
15*	I_{IH1}	GND			5.5 V		GND	GND	GND	GND		GND	GND	2.4 V	GND		40	10X or 4 μA	μA
16*							GND	GND	GND					GND	2.4 V				
17*							2.4 V	GND	GND						GND				
18*							GND	2.4 V	GND										
19*								GND	2.4 V	GND									
20*	I_{IH1}	GND							GND	2.4 V							40	10X or 4 μA	
21	I_{IH2}	5.5 V								GND			GND				100		
22		GND											5.5 V	GND					
23													GND	5.5 V	GND				
24							GND							GND	5.5 V				
25							5.5 V	GND							GND				
26							GND	5.5 V	GND										
27							GND	GND	5.5 V	GND									
28	I_{IH2}	GND					GND	GND	GND	5.5 V			GND	GND	GND		100		μA
29	I_{IL}	0.4 V			5.5 V		5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V	5.5 V	5.5 V	-.7	-1.6		mA

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 02).-Continued.

1/ 2/ Test number	Test name	Pin numbers													Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
30	I_{IL}	5.5 V			5.5 V		5.5 V	5.5 V	5.5 V	5.5 V		GND	0.4 V	5.5 V	5.5 V	-.7	-1.6		mA
31							5.5 V						5.5 V	0.4 V	5.5 V				
32							5.5 V							5.5 V	0.4 V				
33							0.4 V	5.5 V							5.5 V				
34							5.5 V	0.4 V	5.5 V										
35							5.5 V	5.5 V	0.4 V	5.5 V									
36	I_{IL}						5.5 V	5.5 V	5.5 V	0.4 V						-.7	-1.6		
37	I_{CCL}	5.5 V					5.5 V	5.5 V	5.5 V	5.5 V			5.5 V	5.5 V	5.5 V		10		
38	I_{CCH}	GND			5.5 V		GND	GND	GND	GND			GND	GND	GND		3.3		mA
39	V_{IC}	-12 mA			4.5 V												-1.5		V
40													-12 mA						
41														-12 mA					
42															-12 mA				
43	V_{IC}				4.5 V		-12 mA					GND					-1.5		V

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 02). -Continued.

Test number 1/ 2/	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
44	V_{IC}				4.5 V			-12 mA				GND					-1.5		V
45	V_{IC}				4.5 V				-12 mA			GND					-1.5		V
46	V_{IC}				4.5 V					-12 mA		GND					-1.5		V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.

2/ Test sequence is optional with the manufacturer.

3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C. V_{IC} is tested at 25°C only.

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TABLE I. (Device type 03) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	V_{OH}	5.5 V	5.5 V	-.4 mA	4.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V		0.8 V	2.4		±.12	V
2*		0.8 V	5.5 V	-.4 mA			5.5 V								5.5 V				
3*		5.5 V	0.8 V	-.4 mA			5.5 V												
4*			5.5 V			-.4 mA	0.8 V	5.5 V	5.5 V										
5*						-.4 mA	5.5 V	0.8 V	5.5 V										
6*						-.4 mA		5.5 V	0.8 V	5.5 V									
7*									5.5 V	0.8 V	5.5 V			-.4 mA					
8*										5.5 V	0.8 V		5.5 V	-.4 mA					
9*	V_{OH}	5.5 V	5.5 V								5.5 V		0.8 V	-.4 mA	5.5 V	2.4		±.12	
10*	V_{OL}	2.0 V	2.0 V	16 mA			5.5 V	5.5 V	5.5 V		5.5 V		5.5 V		2.0 V		0.4	±.04	
11*	V_{OL}	5.5 V	5.5 V			16 mA	2.0 V	2.0 V	2.0 V	5.5 V	5.5 V		5.5 V		5.5 V		0.4	±.04	
12*	V_{OL}	5.5 V			4.5 V		5.5 V	5.5 V	5.5 V	2.0 V	2.0 V		2.0 V	16 mA	5.5 V		0.4	±.04	V
13*	I_{IL}	5.5 V			5.5 V					5.5 V	5.5 V		5.5 V		0.4 V	-.7	-1.6	±.1	mA
14*		0.4 V	5.5 V												5.5 V	-.7			
15*		5.5 V	0.4 V				5.5 V												
16*			5.5 V				0.4 V	5.5 V											
17*							5.5 V	0.4 V	5.5 V										
18*								5.5 V	0.4 V	5.5 V									
19*									5.5 V	0.4 V	5.5 V								
20*									5.5 V	5.5 V	0.4 V		5.5 V						
21*	I_{IL}	5.5 V	5.5 V				5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		0.4 V		5.5 V	-.7	-1.6	±.1	mA
22*	I_{IH1}	GND	GND				GND	GND	GND	GND	GND		GND		2.4 V		40	10X or 4 µA	µA
23	I_{IH2}	GND	GND												5.5 V		100		
24*	I_{IH1}	2.4 V	GND												GND		40	10X or 4 µA	
25	I_{IH2}	5.5 V	GND														100		
26*	I_{IH1}	GND	2.4 V														40	10X or 4 µA	
27	I_{IH2}		5.5 V				GND										100		
28*	I_{IH1}		GND				2.4 V										40	10X or 4 µA	
29	I_{IH2}	GND	GND		5.5 V		5.5 V	GND	GND	GND	GND	GND	GND		GND		100		µA

See footnotes at end of this table.

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TABLE I. (Device type 03) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
30*	I_{IH1}	GND	GND		5.5 V		GND	2.4 V	GND	GND	GND	GND	GND		GND		40	10X or 4 µA	µA
31	I_{IH2}	↓	↓		↓		↓	5.5 V	GND	↓	↓	↓	↓		↓		100		↓
32*	I_{IH1}	↓	↓		↓		↓	GND	2.4 V	↓	↓	↓	↓		↓		40	10X or 4 µA	↓
33	I_{IH2}	↓	↓		↓		↓	↓	5.5 V	GND	↓	↓	↓		↓		100		↓
34*	I_{IH1}	↓	↓		↓		↓	↓	GND	2.4 V	↓	↓	↓		↓		40	10X or 4 µA	↓
35	I_{IH2}	↓	↓		↓		↓	↓	↓	5.5 V	GND	↓	↓		↓		100		↓
36*	I_{IH1}	↓	↓		↓		↓	↓	↓	GND	2.4 V	↓	↓		↓		40	10X or 4 µA	↓
37	I_{IH2}	↓	↓		↓		↓	↓	↓	GND	5.5 V	↓	GND		↓		100		↓
38*	I_{IH1}	↓	↓		↓		↓	↓	↓	GND	GND	↓	2.4 V		↓		40	10X or 4 µA	↓
39	I_{IH2}	↓	↓		↓		GND	GND	GND	GND	GND	↓	5.5 V		↓		100		µA
40	I_{OS}	GND	GND	GND	↓		↓	↓	↓	↓	↓	↓	↓		GND	-20	-55		mA
41	I_{OS}				↓	GND	GND	GND	GND			↓	↓		↓	-20	-55		mA
42	I_{OS}				5.5 V		↓	↓	↓	GND	GND	↓	GND	0 V	↓	-20	-55		mA
43	V_{IC}	-12 mA			4.5 V		↓	↓	↓	↓	↓	↓	↓	↓	↓		-1.5		V
44	↓	↓	-12 mA		↓		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
45	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	-12 mA	↓	↓	↓	↓
46	↓	↓	↓	↓	↓	↓	-12 mA	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
47	↓	↓	↓	↓	↓	↓	↓	-12 mA	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
48	↓	↓	↓	↓	↓	↓	↓	↓	-12 mA	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
49	↓	↓	↓	↓	↓	↓	↓	↓	↓	-12 mA	↓	↓	↓	↓	↓	↓	↓	↓	↓
50	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	-12 mA	↓	↓	↓	↓	↓	↓	↓	↓
51	V_{IC}	↓	↓	↓	4.5 V	↓	↓	↓	↓	↓	↓	↓	-12 mA	↓	↓	↓	-1.5	↓	V
52	I_{CCH}	GND	GND		5.5 V		GND	GND	GND	GND	GND	↓	GND		GND		4.95		mA
53	I_{CCL}	5.5 V	5.5 V		5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V		5.5 V		15		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with delta read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C. V_{IC} tests are omitted at temperature extremes.

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TABLE I. Electrical characteristics test (device type 04).

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
1*	V_{OL}	2.0 V	2.0 V	16 mA	4.5 V		5.5 V	5.5 V		5.5 V	5.5 V	GND	5.5 V	5.5 V			.4	±.04	V
2*	↓	5.5 V	5.5 V		↓	16 mA	2.0 V	2.0 V		5.5 V	5.5 V	↓	5.5 V	5.5 V			↓	↓	↓
3*	↓	5.5 V	5.5 V		↓	↓	5.5 V	5.5 V	16 mA	2.0 V	2.0 V	↓	5.5 V	5.5 V			↓	↓	↓
4*	V_{OL}	5.5 V	5.5 V		↓	↓	5.5 V	5.5 V		5.5 V	5.5 V	↓	2.0 V	2.0 V	16 mA		.4	±.04	↓
5*	V_{OH}	0.8 V	5.5 V	-.4 mA	↓	↓	5.5 V	5.5 V		5.5 V	↓	↓	5.5 V	5.5 V		2.4		±.120	↓
6*	↓	5.5 V	0.8 V	-.4 mA	↓	↓	5.5 V	5.5 V		5.5 V	↓	↓	↓	↓	↓	↓	↓	↓	↓
7*	↓	↓	5.5 V	↓	↓	-.4 mA	0.8 V	5.5 V		5.5 V	↓	↓	↓	↓	↓	↓	↓	↓	↓
8*	↓	↓	↓	↓	↓	-.4 mA	5.5 V	0.8 V		5.5 V	↓	↓	↓	↓	↓	↓	↓	↓	↓
9*	↓	↓	↓	↓	↓	↓	↓	5.5 V	-.4 mA	0.8 V	5.5 V	↓	↓	↓	↓	↓	↓	↓	↓
10*	↓	↓	↓	↓	↓	↓	↓	↓	-.4 mA	5.5 V	0.8 V	↓	5.5 V	↓	↓	↓	↓	↓	↓
11*	↓	↓	↓	↓	↓	↓	↓	↓	↓	5.5 V	5.5 V	↓	0.8 V	5.5 V	-.4 mA	↓	↓	↓	↓
12*	V_{OH}	5.5 V	5.5 V		4.5 V		5.5 V	5.5 V		5.5 V	5.5 V	↓	5.5 V	0.8 V	-.4 mA	2.4		±.120	V
13	I_{OS}	GND	GND	GND	5.5 V							↓				-20	-55		mA
14	I_{OS}				5.5 V	GND	GND	GND				GND				-20	-55		mA

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 04). -Continued

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
15	I_{OS}				5.5 V				GND	GND	GND	GND				-20	-55		mA
16	I_{OS}												GND	GND	GND	-20	-55		mA
17*	I_{IH1}	2.4 V	GND				GND	GND		GND	GND						40	10X or 4 μA	μA
18*		GND	2.4 V				GND	GND		GND									
19*			GND				2.4 V	GND		GND									
20*							GND	2.4 V		GND									
21*							GND	GND		2.4 V	GND								
22*										GND	2.4 V		GND						
23*											GND		2.4 V	GND					
24*	I_{IH1}	GND											GND	2.4 V			40	10X or 4 μA	
25	I_{IH2}	5.5 V	GND											GND			100		
26		GND	5.5 V				GND												
27			GND				5.5 V	GND											
28			GND				GND	5.5 V		GND									
29	I_{IH2}	GND	GND		5.5 V		GND	GND		5.5 V	GND	GND	GND	GND			100		μA

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 04). -Continued

1/ 2/ Test number	Test name	Pin numbers													Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
30	I_{IH2}	GND	GND		5.5 V		GND	GND		GND	5.5 V	GND	GND	GND			100		µA
31	I_{IH2}	GND	GND				GND	GND		GND	GND		5.5 V	GND			100		µA
32	I_{IH2}	GND	GND				GND	GND		GND	GND		GND	5.5 V			100		µA
33	I_{IL}	0.4 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	5.5 V		-.7	-1.6		mA
34		5.5 V	0.4 V				5.5 V	5.5 V											
35			5.5 V				0.4 V	5.5 V											
36							5.5 V	0.4 V		5.5 V									
37								5.5 V		0.4 V	5.5 V								
38										5.5 V	0.4 V		5.5 V						
39										5.5 V	5.5 V		0.4 V	5.5 V					
40	I_{IL}	5.5 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	0.4 V		-.7	-1.6		
41	I_{CCH}	GND	GND				GND	GND		GND	GND		GND	GND			6.6		
42	I_{CCL}	5.5 V	5.5 V		5.5 V		5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	5.5 V			20		mA
43	V_{IC}	-12 mA			4.5 V							GND					-1.5		V

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 04). -Continued

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
44	V_{IC}		-12 mA		4.5 V							GND					-1.5		V
45							-12 mA												
46								-12 mA											
47										-12 mA									
48											-12 mA								
49													-12 mA						
50	V_{IC}				4.5 V							GND		-12 mA			-1.5		V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C. V_{IC} is tested at 25°C only.

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TABLE I. Electrical characteristics test (device type 05).

1/2/ Test number	Test name	Pin numbers													Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
1*	V_{OL}	2.0 V		5.5 V	4.5 V	5.5 V		5.5 V		5.5 V		GND		5.5 V	16 mA		0.4	±.04	V
2*		5.5 V	16 mA	2.0 V		5.5 V		5.5 V											
3*				5.5 V		2.0 V	16 mA	5.5 V											
4*						5.5 V		2.0 V	16 mA	5.5 V									
5*								5.5 V		2.0 V	16 mA			5.5 V					
6*	V_{OL}	5.5 V								5.5 V			16 mA	2.0 V			0.4	±.04	
7*	V_{OH}	0.8 V		5.5 V										5.5 V	-.4 mA	2.4		±.12	
8*		5.5 V	-.4 mA	0.8 V		5.5 V													
9*				5.5 V		0.8 V	-.4 mA	5.5 V											
10*						5.5 V		0.8 V	-.4 mA	5.5 V									
11*						5.5 V		5.5 V		0.8 V	-.4 mA			5.5 V					
12*	V_{OH}	5.5 V		5.5 V	4.5 V	5.5 V		5.5 V		5.5 V			-.4 mA	0.8 V		2.4		±.12	V
13	I_{OS}	GND			5.5 V										GND	-20	-55		mA
14	I_{OS}		GND	GND	5.5 V							GND				-20	-55		mA

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 05). -Continued.

1/2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
15	I_{OS}				5.5 V	GND	GND					GND				-20	-55		mA
16	↓							GND	GND							↓	↓		↓
17	↓									GND	GND					↓	↓		↓
18	I_{OS}												GND	GND		-20	-55		mA
19*	I_{IH1}	2.4 V		GND		GND		GND		GND				GND			40	10X or 4 μA	μA
20*	↓	GND		2.4 V		GND		GND		GND				GND			↓	↓	↓
21*	↓	GND		GND		2.4 V		GND		GND				GND			↓	↓	↓
22*	↓	↓		↓		GND		2.4 V		GND				GND			↓	↓	↓
23*	↓	↓		↓		↓		GND		2.4 V				GND			↓	↓	↓
24*	I_{IH1}	GND		↓		↓		↓		GND				2.4 V			40	10X or 4 μA	↓
25	I_{IH2}	5.5 V		GND		↓		↓		↓				GND			100		↓
26	↓	GND		5.5 V		GND		↓		↓		↓		↓			↓		↓
27	↓			GND	↓	5.5 V		GND		↓		↓		↓			↓		↓
28	I_{IH2}	GND		GND	5.5 V	GND		5.5 V		GND		GND		GND			100		μA

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TABLE I. Electrical characteristics test (device type 05). -Continued.

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
29	I_{IH2}	GND		GND	5.5 V	GND		GND		5.5 V		GND		GND			100		µA
30	I_{IH2}	GND		GND	↓	GND		GND		GND		↓		5.5 V			100		µA
31	I_{IL}	0.4 V		5.5 V	↓	5.5 V		5.5 V		5.5 V		↓		↓		-.7	-1.6		mA
32	↓	5.5 V		0.4 V	↓	5.5 V		5.5 V		↓		↓		↓		↓	↓		↓
33	↓	↓		5.5 V	↓	0.4 V		5.5 V		↓		↓		↓		↓	↓		↓
34	↓	↓		↓	↓	5.5 V		0.4 V		5.5 V		↓		↓		↓	↓		↓
35	↓	↓		↓	↓	5.5 V		5.5 V		0.4 V		↓		5.5 V		↓	↓		↓
36	I_{IL}	5.5 V		5.5 V	↓	5.5 V		5.5 V		5.5 V		↓		0.4 V		-.7	-1.6		↓
37	I_{CCL}	5.5 V		5.5 V	↓	5.5 V		5.5 V		5.5 V		↓		5.5 V			30		↓
38	I_{CCH}	GND		GND	5.5 V	GND		GND		GND		↓		GND			9.9		mA
39	V_{IC}	-12 mA			4.5 V							↓					-1.5		V
40	V_{IC}			-12 mA	4.5 V							↓					-1.5		V
41	V_{IC}				4.5 V	-12 mA						↓					-1.5		V
42	V_{IC}				4.5 V			-12 mA				GND					-1.5		V

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TABLE I. Electrical characteristics test (device type 05).-Continued.

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
43	V_{IC}				4.5 V					-12 mA		GND					-1.5		V
44	V_{IC}				4.5 V							GND		-12 mA			-1.5		V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded. ("10X" denotes "10 times the initial measurement".) Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C. V_{IC} is tested at 25°C only.

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TABLE I. Electrical characteristics tests (device type 06).

Test No. 1/2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2	-	-	V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V	-	0.8	-	V
3	Maximum collector cut-off current	I_{CEX}	V_{CC} = 4.5 V, V_{OH} = 5.5 V V_{IL} = 0.8 V	-	250	-	µA
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V	-	0.4	±.04	V
5*	High level input current (each input)	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.4 V	-	40	±4.0	µA
		I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V	-	100	N/A	µA
6	Low level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IL} = 0.4 V	-0.7	-1.6	-	mA
7	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_A = 25°C	-	-1.5		V
8	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IH} = 5.5 V	-	16.5	-	mA
9	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IH} = 4.5 V V_{IL} = 0 V	-	6	-	mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. Electrical characteristics test (device type 07).

1/ 2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
1*	V_{OL}	2.0 V	2.0 V	16 mA	4.5 V		5.5 V	5.5 V		5.5 V	5.5 V	GND	5.5 V	5.5 V			0.4	±.04	V
2*	↓	5.5 V	5.5 V			16 mA	2.0 V	2.0 V		5.5 V	5.5 V		5.5 V	5.5 V			↓	↓	↓
3*	↓	5.5 V	5.5 V				5.5 V	5.5 V	16 mA	2.0 V	2.0 V		5.5 V	5.5 V			↓	↓	↓
4*	V_{OL}	5.5 V	5.5 V				↓	↓		5.5 V	5.5 V		2.0 V	2.0 V	16 mA		0.4	±.04	V
5*	I_{CEX}	0.8 V	4.5 V	5.5 V			↓	↓		↓	↓		5.5 V	5.5 V			250	10X or 25 μA	μA
6*	↓	4.5 V	0.8 V	5.5 V			5.5 V	5.5 V					↓	↓			↓	↓	↓
7*	↓	5.5 V	5.5 V			5.5 V	0.8 V	4.5 V		↓	↓								
8*	↓	↓	↓			5.5 V	4.5 V	0.8 V		5.5 V	5.5 V								
9*	↓	↓	↓				5.5 V	5.5 V	5.5 V	0.8 V	4.5 V		↓	↓					
10*	↓	↓	↓		↓		↓	↓	5.5 V	4.5 V	0.8 V		5.5 V	5.5 V			↓	↓	↓
11*	↓	↓	↓		↓		↓	↓		5.5 V	5.5 V		0.8 V	4.5 V	5.5 V		↓	↓	↓
12*	I_{CEX}	5.5 V	5.5 V		4.5 V		5.5 V	5.5 V		5.5 V	5.5 V		4.5 V	0.8 V	5.5 V		250	10X or 25 μA	
13*	I_{IH1}	2.4 V	GND		5.5 V		GND	GND		GND	GND		GND	GND			40	10X or 4 μA	
14*	↓	GND	2.4 V		↓		GND	GND		↓	↓	↓	↓	↓			↓	↓	↓
15*	↓	GND	GND		↓		2.4 V	GND		↓	↓	↓	↓	↓			↓	↓	↓
16*	I_{IH1}	GND	GND		5.5 V		GND	2.4 V		GND	GND	GND	GND	GND			40	10X or 4 μA	μA

See footnotes at end of this table.

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TABLE I. Electrical characteristics test (device type 07). -Continued.

1/2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
17*	I_{IH1}	GND	GND		5.5 V		GND	GND		2.4 V	GND	GND	GND	GND			40	10X or 4 μA	μA
18*	I_{IH1}	GND	GND							GND	2.4 V		GND	GND			40	10X or 4 μA	
19*	I_{IH1}	GND	GND								GND		2.4 V	GND			40	10X or 4 μA	
20*	I_{IH1}	GND	GND										GND	2.4 V			40	10X or 4 μA	
21	I_{IH2}	5.5 V	GND											GND			100		
22		GND	5.5 V				GND												
23			GND				5.5 V	GND											
24							GND	5.5 V		GND									
25								GND		5.5 V	GND								
26										GND	5.5 V		GND						
27										GND	GND		5.5 V	GND					
28	I_{IH2}	GND	GND				GND	GND		GND	GND		GND	5.5 V			100		μA
29	I_{IL}	0.4 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V			-.7	-1.6		mA
30		5.5 V	0.4 V				5.5 V	5.5 V											
31			5.5 V				0.4 V	5.5 V											
32			5.5 V				5.5 V	0.4 V		5.5 V									
33	I_{IL}	5.5 V	5.5 V		5.5 V		5.5 V	5.5 V		0.4 V	5.5 V	GND	5.5 V	5.5 V		-.7	-1.6		mA

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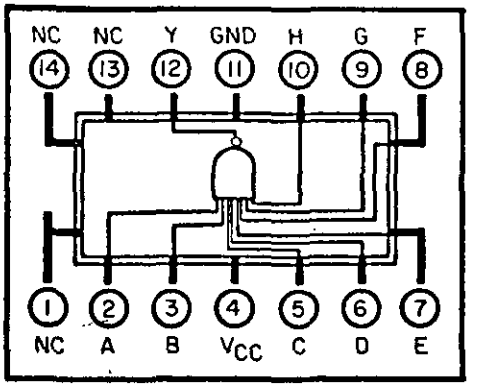
TABLE I. Electrical characteristics test (device type 07). -Continued.

1/2/ Test number	Test name	Pin numbers														Limits 3/		Delta 1/ Δ	Units
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	Min	Max		
34	I_{IL}	5.5 V	5.5 V		5.5 V		5.5 V	5.5 V		5.5 V	0.4 V	GND	5.5 V	5.5 V		-.7	-1.6		mA
35	I_{IL}	5.5 V	5.5 V		5.5 V		5.5 V	5.5 V		5.5 V	5.5 V		0.4 V	5.5 V		-.7	-1.6		mA
36	I_{IL}	5.5 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	0.4 V		-.7	-1.6		mA
37	I_{CCL}	5.5 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	5.5 V			20		mA
38	I_{CCH}	GND	GND		5.5 V		GND	GND		GND	GND		GND	GND			6.6		mA
39	V_{IC}	-12 mA			4.5 V												-1.5		V
40			-12 mA																
41							-12 mA												
42								-12 mA											
43										-12 mA									
44											-12 mA								
45													-12 mA						
46	V_{IC}				4.5 V							GND		-12 mA			-1.5		V

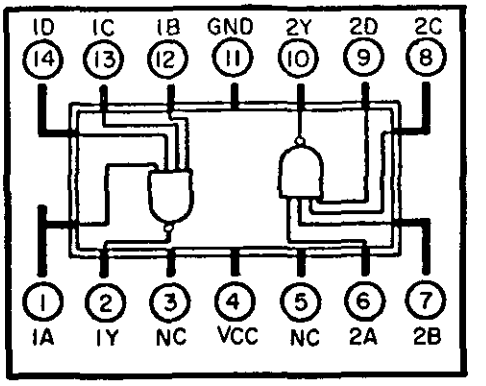
1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C. V_{IC} is tested at 25°C only.

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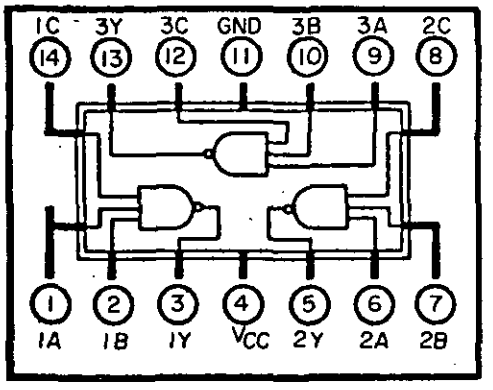
DEVICE TYPE 01



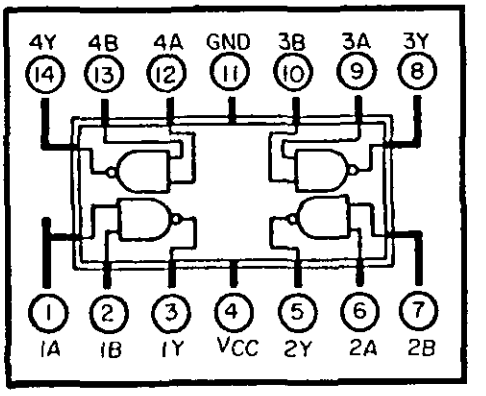
DEVICE TYPE 02



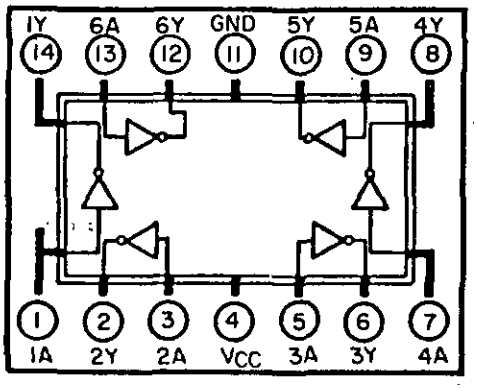
DEVICE TYPE 03



DEVICE TYPES 04 AND 07



DEVICE TYPE 05



DEVICE TYPE 06

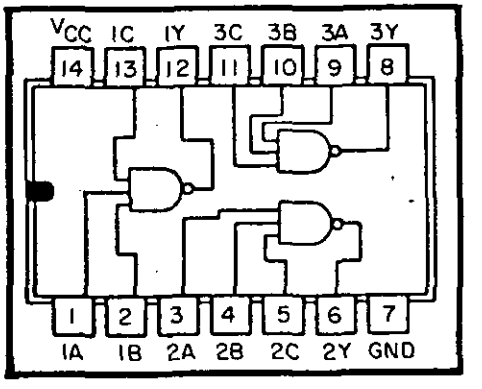


FIGURE 1. Logic diagrams and terminal connections.

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Device type 01

Truth table								
Input								Output
A	B	C	D	E	F	G	H	Y
H	H	H	H	H	H	H	H	L
All other combinations of H and L at the inputs give H output.								

Positive logic $Y = \overline{ABCDEFGH}$

Device type 05

Truth table each gate	
Input	Output
A	Y
L	H
H	L

Positive logic $Y = \overline{A}$

Device types 03 and 06

Truth table			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positive logic $Y = \overline{ABC}$

Device type 02

Truth table				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

Device types 04 and 07

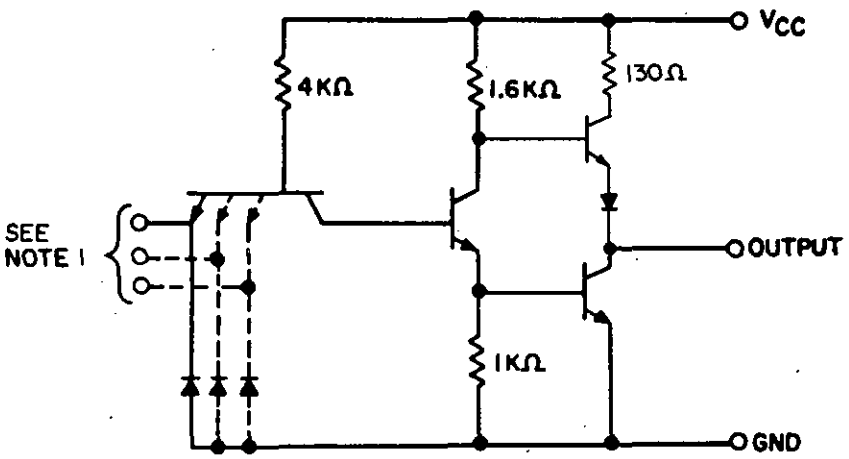
Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive logic $Y = \overline{AB}$

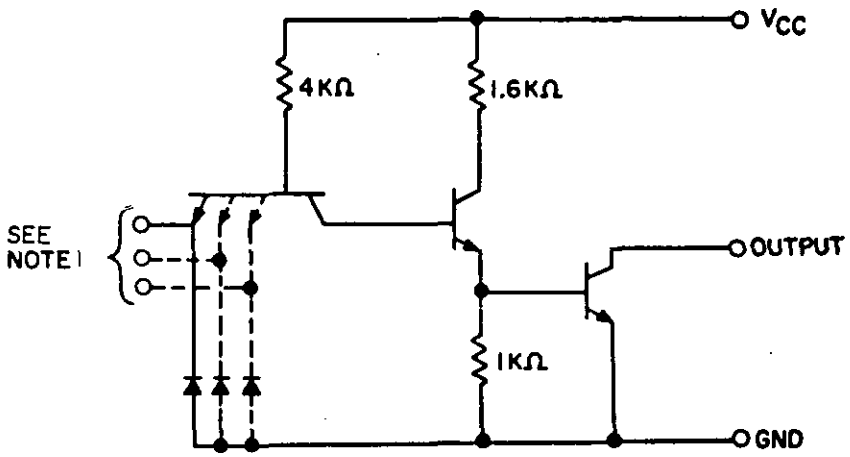
FIGURE 2. Truth tables and logic equations.

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Device types 01, 02, 03, 04 and 05



Device types 06 and 07

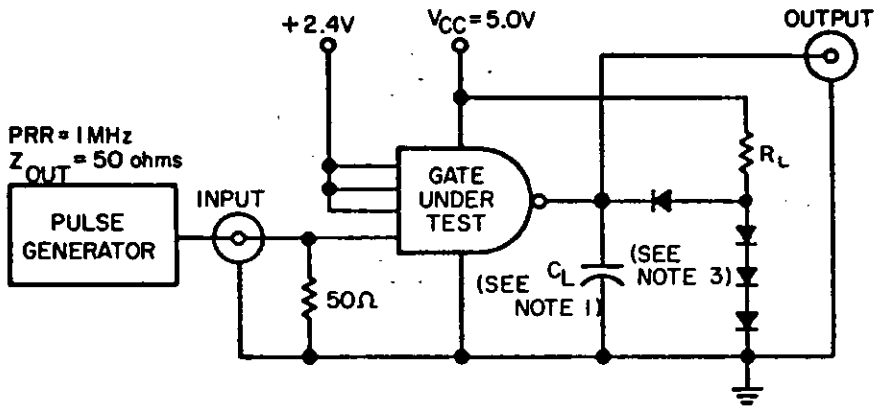


NOTES:
1. Number of inputs is determined by logic configuration shown in figure 1.
2. All resistance values shown are nominal.

FIGURE 3. Schematic circuits.

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Device types 01, 02, 03, 04 and 05



Device types 06 and 07

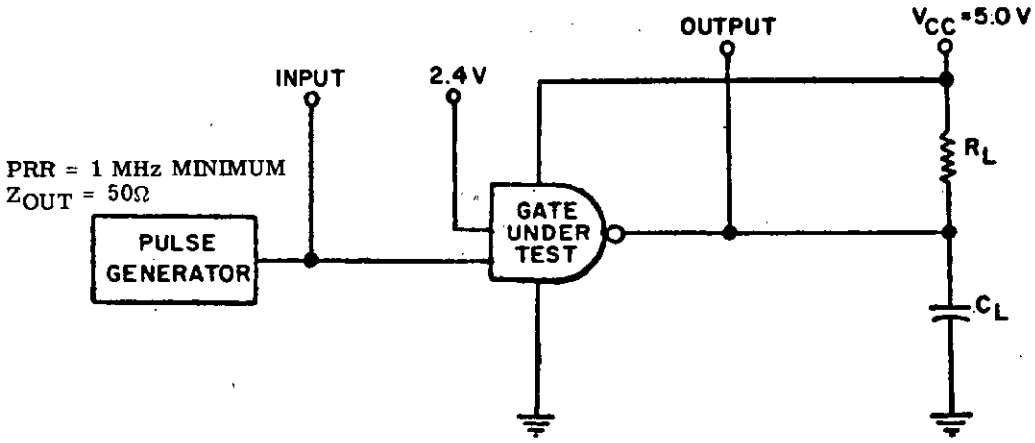
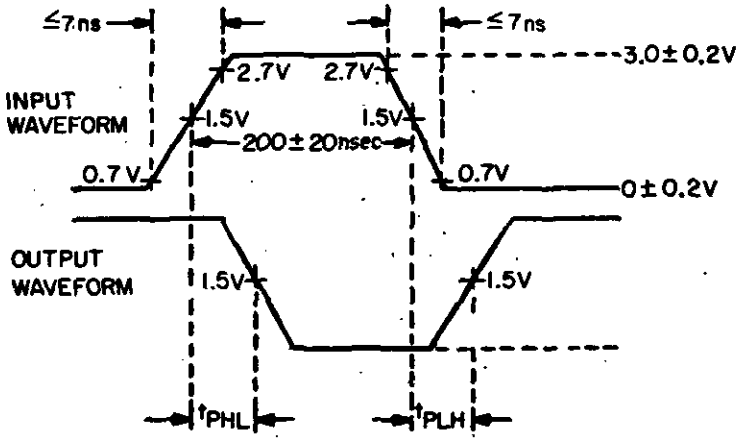


FIGURE 4. Switching time test circuit.

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NOTES:

1. C_L = 50 pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 390 ohms ±5%.
5. Generator pulse width is 500 ns for device type 03.
6. $t_{pd} = \frac{t_{PHL} - t_{PLH}}{2}$

FIGURE 4. Switching time test circuit - Continued.

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TABLE II. Switching characteristics (device types 01, 02, 03, 04, and 05).

Characteristic 1/	Symbol	Conditions	Test figure	Temp.	Limits Min	Limits Max	Units
Propagation delay time high to low level	t_{PHL}	C_L = 50 pF R_L = 390Ω ±5% V_{CC} = 5 V	4	25°C	3	20	ns
				125°C, -55°C	3	24	
Propagation delay time low to high level	t_{PLH}		4	25°C	3	25	ns
				125°C, -55°C	3	27	

1/ Test each input gate.

TABLE II. Switching characteristics (device types 06 and 07).

Characteristic 1/	Symbol	Conditions	Test figure	Temp.	Limits Min	Limits Max	Units
Propagation delay time high to low level	t_{PHL}	C_L = 50 pF R_L = 390Ω ±5% V_{CC} = 5 V	4	25°C	3	23	ns
				125°C, -55°C	3	29	
Propagation delay time low to high level	t_{PLH}		4	25°C	3	28	ns
				125°C, -55°C	3	35	

1/ Test each input gate.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 shall be performed at 25°C. All devices shall meet the limits specified in table II.

3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. Quality assurance shall be in accordance with MIL-M-0038510.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19, 85
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-51)

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